Reading a research claim

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

One complete claim and supporting-evidence audit.

Keep headline, outcome, unit, data, baseline, arithmetic and unsupported extensions together.

Keep one habit: Recall → W1–W3/W5 → independent median claim → research-claim project → delayed return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from A question that **evidence** can answer. Give an example and a mistake that the idea helps you avoid.

A question that evidence can answer

> **My remembered idea, example and correction**
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MAIN LESSON · SECTION 5

W1 · Explain the current example

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Reading a research claim requires its method, data, baseline, metric, unit and limitations. Main fictional same-eight-case supported-answer counts A 5/8 = 62.5%, B 6/8 = 75%: +12.5 percentage points, rate ratio 1.2 and relative increase 20%. This supports a descriptive measured-outcome comparison, not universal usefulness, causal mechanism or learner benefit.

  Actual paired transitions, full support rubric and uncertainty analysis are not supplied by the headline. Added possible paired patterns illustrate that identical aggregates can contain one gain/no loss or two gains/one loss; they are not recovered real records.

  Separate picture correct-answer rates 2/10 = 20% to 4/10 = 40%: twice the rate, +20 percentage points and 100% relative increase, with six failures remaining. “Every learner improves twice as much” changes unit/outcome without data. No significance, population interval, causal effect or actual paper/model/participant result is supplied. A zero baseline has undefined finite rate ratio.

  Keep a valid narrow result while revising an unsupported headline.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

MAIN LESSON · SECTION 6

W2 · Trace the worked case

> Keep the supplied inputs, comparison rule and evidence together.

Complete task and evidence

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  Keep a valid narrow result while revising an unsupported headline.

- A fictional report says method B is better after 6/8 supported answers versus baseline 5/8 on the same constructed cases. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task's inputs,rule and response together.

Complete task and evidence

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  Separate picture correct-answer rates 2/10 = 20% to 4/10 = 40%: twice the rate, +20 percentage points and 100% relative increase, with six failures remaining. “Every learner improves twice as much” changes unit/outcome without data. No significance, population interval, causal effect or actual paper/model/participant result is supplied. A zero baseline has undefined finite rate ratio.

  Keep a valid narrow result while revising an unsupported headline.

- A report omits this qualification: A numerical improvement can be real in the sample while the broader **claim** remains unsupported.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

W5 · Explain the limitation

> Keep this task’s inputs,rule and response together.

Complete task and evidence

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  Actual paired transitions, full support rubric and uncertainty analysis are not supplied by the headline. Added possible paired patterns illustrate that identical aggregates can contain one gain/no loss or two gains/one loss; they are not recovered real records.

  Separate picture correct-answer rates 2/10 = 20% to 4/10 = 40%: twice the rate, +20 percentage points and 100% relative increase, with six failures remaining. “Every learner improves twice as much” changes unit/outcome without data. No significance, population interval, causal effect or actual paper/model/participant result is supplied. A zero baseline has undefined finite rate ratio.

  Keep a valid narrow result while revising an unsupported headline.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Report **claim**s universal speed after median 2 seconds versus 3 on four local fixtures. Audit scope.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply a new case independently

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 10

Project · Keep the research evidence

> Preserve the question, protocol, observations, analysis and limits.

Complete task and evidence

- Add a labelled reading a research **claim evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of reading a research **claim**.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	

Evidence	My record
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Trace the worked case for Reading a research **claim**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for Reading a research **claim**. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Report **claim**s universal speed after median 2 seconds versus 3 on four local fixtures. Audit scope.

My reasoning and evidence

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

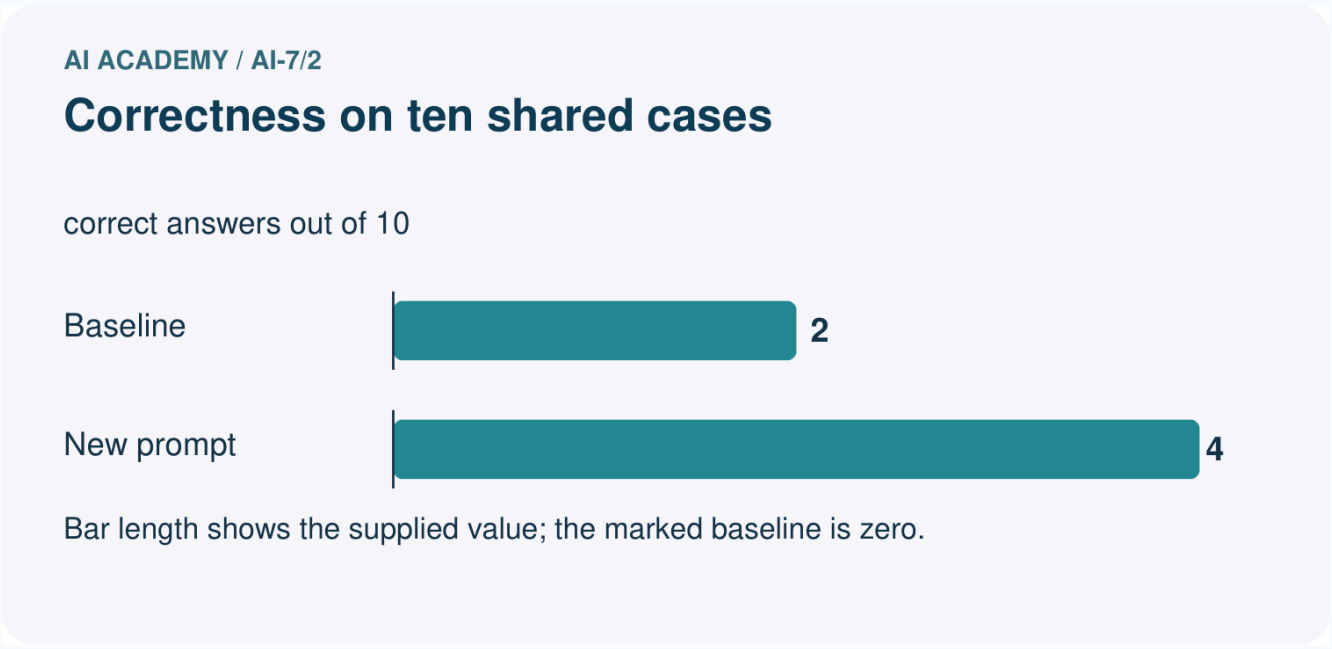
> Keep this task's inputs,rule and response together.

Complete task and evidence

- Read a **claim** with its denominator. A fictional paper headline says a prompt is twice as effective. Its table reports four correct cases instead of two.
- Baseline: 2/10 correct. New prompt: 4/10 on the same ten cases. Scoring is fixed.
- Calculate the baseline and new rates for 2/10 and 4/10. Explain both twice the rate and the percentage-point change, and say why neither measures every learner's improvement.
- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning.

  Connect the **claim** to its records, units and assumptions so another investigator can challenge or reproduce it.



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/2

Reading a research claim

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Find the measured outcome

The table counts correct answers, not learning or satisfaction.



2. Calculate both changes

20% -> 40% is twice the rate and a 20-percentage-point increase.



3. Check claim scope

These ten cases support a descriptive comparison, not universal doubling.

CHANGE ONE THING

The headline changes to "every learner improves twice as much".

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Project: Reproducible AI Research Study
- Current milestone: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.
- For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the **claim** to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Read a **claim** with its denominator. A fictional paper headline says a prompt is twice as effective. Its table reports four correct cases instead of two.
- Which labelled step matches this detail? These ten cases support a descriptive comparison, not universal doubling.
- 1. Check **claim** scope
- 2. Calculate both changes
- 3. Find the measured outcome
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Check **claim** scope These ten cases support a descriptive comparison, not universal doubling. A research **claim** compresses a method and result.

Unpack the measured variable, denominator and comparison before accepting the headline.

Changed-case question and optional hint

- Changed-case practice: The headline changes to "every learner improves twice as much". Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. This one example guarantees correct results for every future **input**.
- 3. That learner **claim** is unsupported by these answer counts.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ Check after the changed attempt

After your attempt, compare with the supported explanation: That learner **claim** is unsupported by these answer counts. The outcome and experimental unit have changed in the wording without new **evidence**.

Repair and retain the record

Review A question that **evidence** can answer (ai-7/1). Goal: Explain the **mechanism** and terminology of a question that **evidence** can answer. Short practice: Which proposed diagram question has a measurable outcome?

Name its versions, six-case denominator and fixed scoring, then explain why giving only B easier tasks spoils the comparison.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
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▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

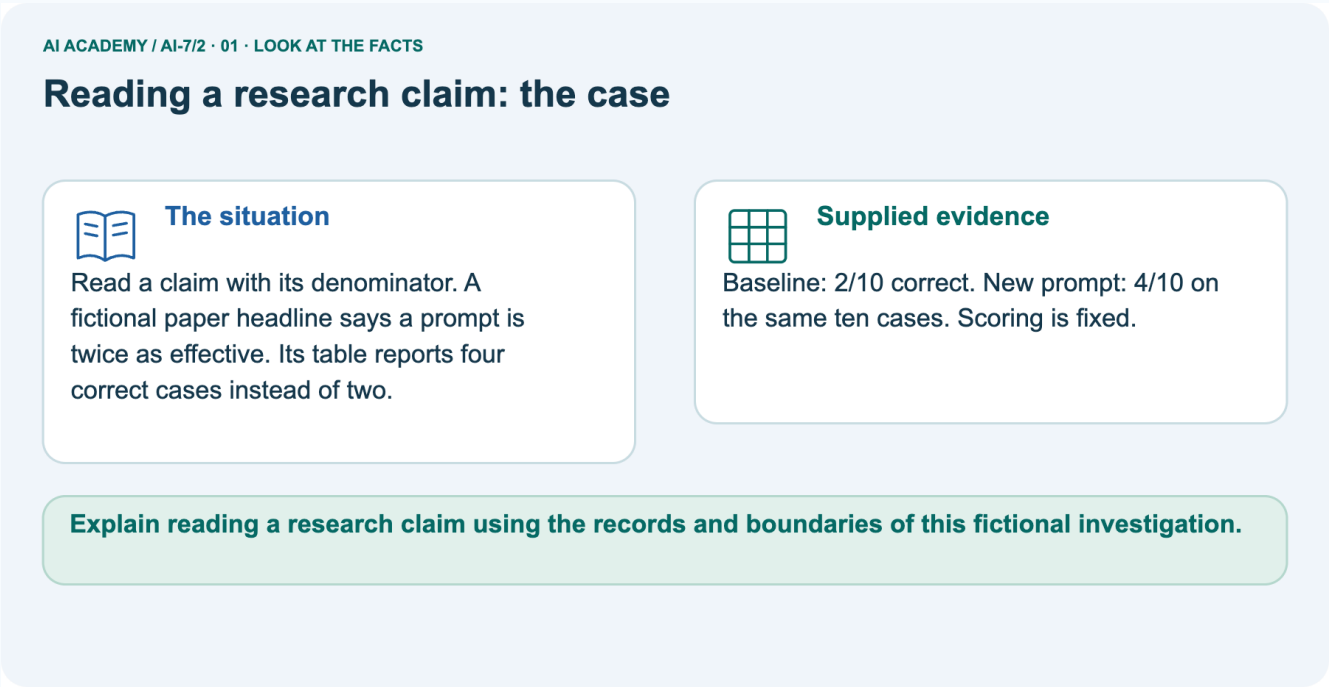
▼ Reference · Six-picture response sheets

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Reading a research claim: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

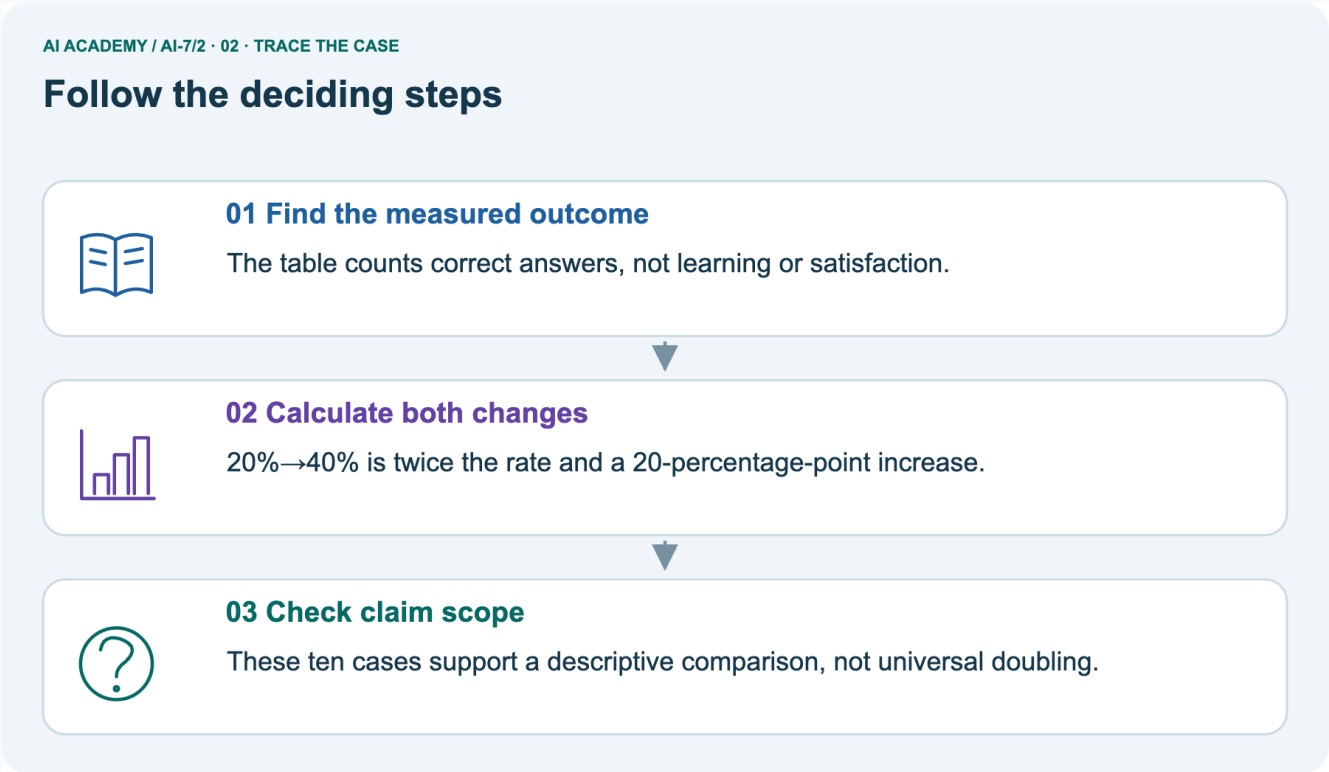
Read a **claim** with its denominator. A fictional paper headline says a prompt is twice as effective. Its table reports four correct cases instead of two. Supplied facts: Baseline: 2/10 correct.

New prompt: 4/10 on the same ten cases. Scoring is fixed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



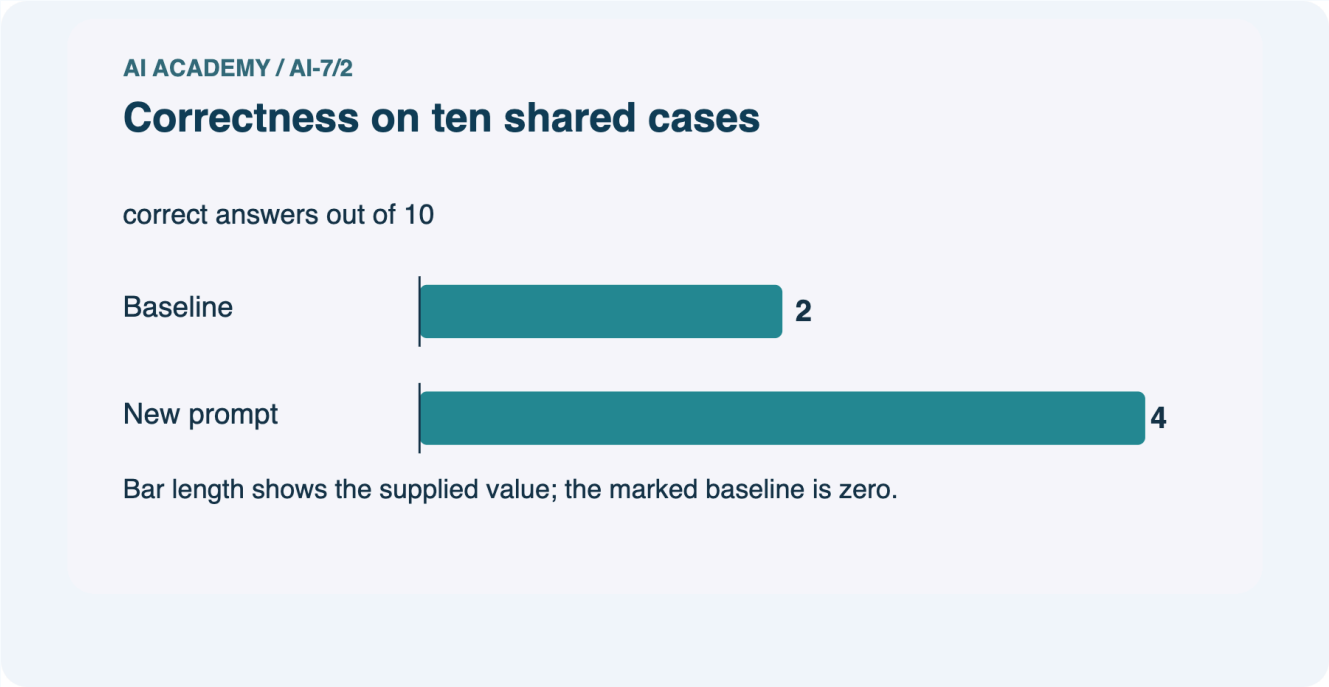
Follow the labelled picture with the matching explanation. Open the vector diagram

Find the measured outcome: The table counts correct answers, not learning or satisfaction.
→ Calculate both changes: 20%→40% is twice the rate and a 20-percentage-point increase.
→ Check **claim** scope: These ten cases support a descriptive comparison, not universal doubling.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Correctness on ten shared cases



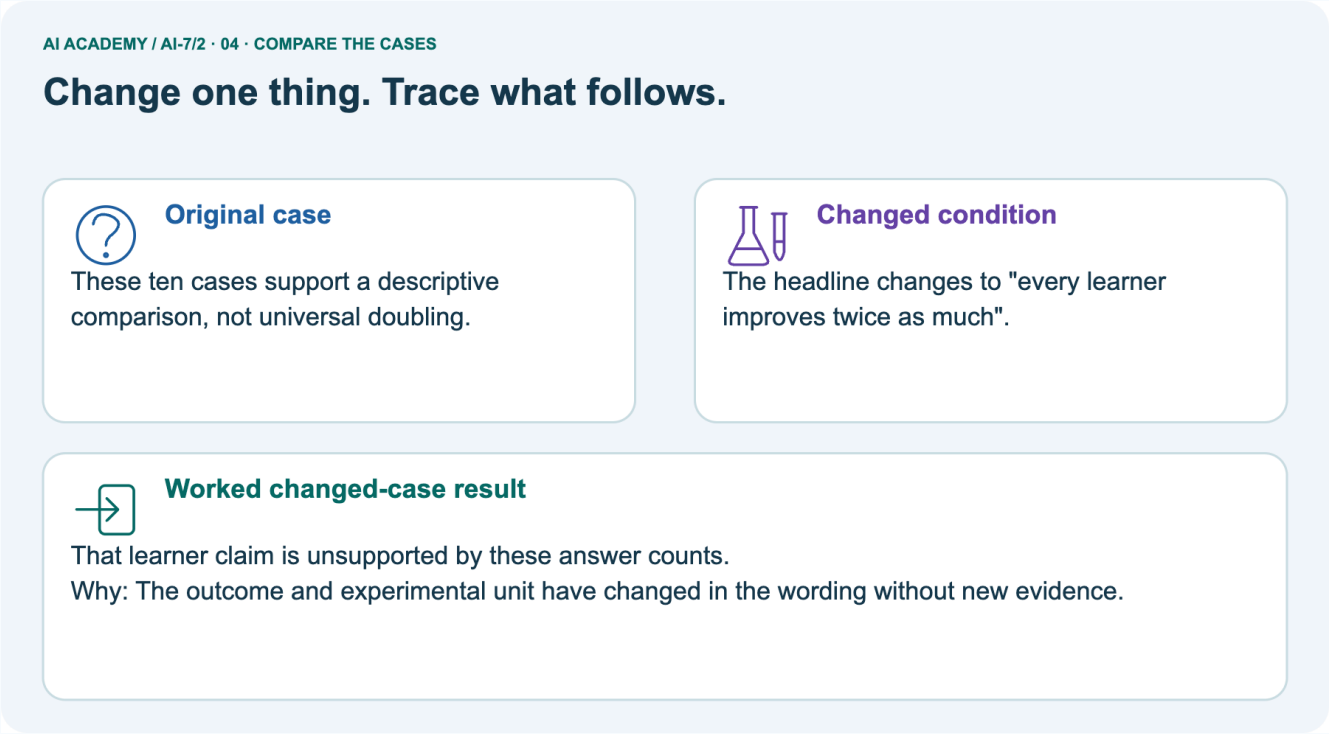
Follow the labelled picture with the matching explanation. Open the vector diagram

Correctness on ten shared cases. Units: correct answers out of 10. Baseline: 2. New prompt: 4.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

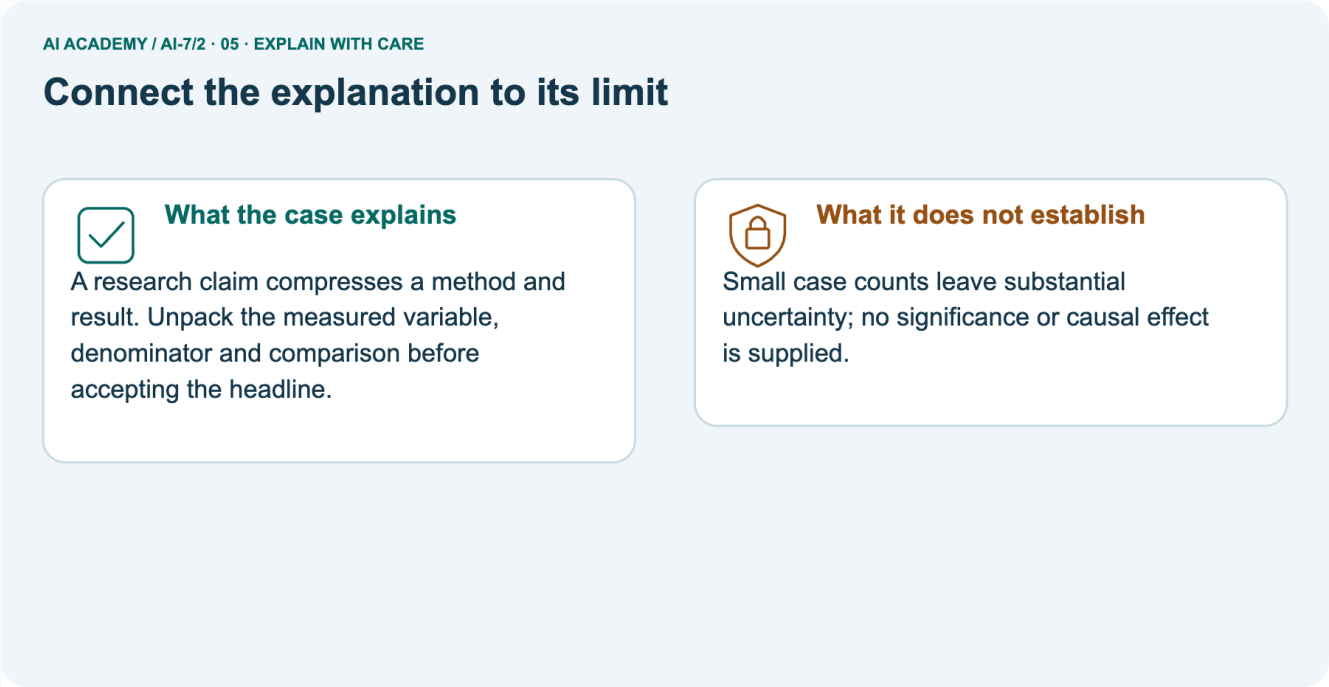
Original case: These ten cases support a descriptive comparison, not universal doubling.
Changed condition: The headline changes to "every learner improves twice as much".

Expected result: That learner **claim** is unsupported by these answer counts. Reason: The outcome and experimental unit have changed in the wording without new **evidence**.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

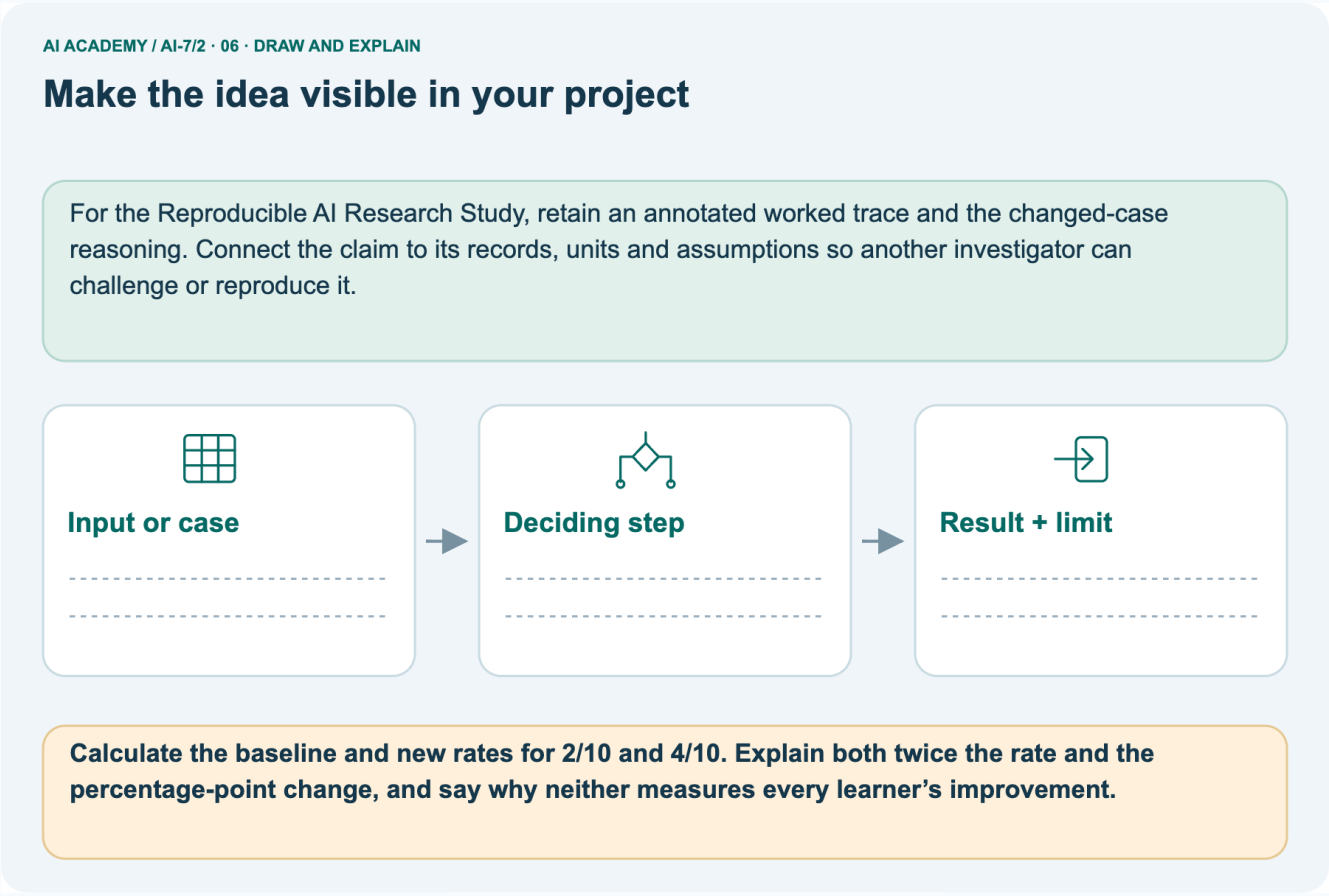
A research **claim** compresses a method and result. Unpack the measured variable, denominator and comparison before accepting the headline.

Limit: Small case counts leave substantial uncertainty; no significance or causal effect is supplied.

Explain the case, then point to a limit on the **claim**.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the **claim** to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Calculate the baseline and new rates for 2/10 and 4/10.

Explain both twice the rate and the percentage-point change, and say why neither measures every learner's improvement.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Reading a research claim: the case

02. Follow the deciding steps

03. Correctness on ten shared cases

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.